Multivariate Normal Distribution

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Outline

- The multivariate normal distribution
- The Bivariate Normal Distribution
- More properties of multivariate normal
- Multivariate Normality and Outliers
- Estimation of μ and Σ
- Central Limit Theorem

Motivation

- To be able to make inferences about populations, we need a model for the distribution of random variables, We'll use the multivariate normal distribution, because...
- It's often a good population model. It's a reasonably good approximation of many phenomenon. A lot of variables are approximately normal (due to the central limit theorem for sums and averages).
- The sampling distribution of (test) statistics are often approximately multivariate or univariate normal due to the central limit theorem
- Due to it's central importance, we need to thoroughly understand and know it's properties.

Introduction to the Multivariate Normal

- The probability density function of the Univariate normal distribution ($p = 1$ variables):

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\}, \quad \text{for} \quad -\infty < x < \infty$$

- The parameters that completely characterize the distribution:
  1. $\mu = E(X) =$ mean.
  2. $\sigma^2 = var(X) =$ variance
- Area corresponds to probability:
  68% area between $\mu \pm \sigma$ and 95% between $\mu \pm 1.96\sigma$
- As shorthand notation we may use the expression below:

$$X \sim N(\mu, \sigma^2)$$

Generalization to Multivariate Normal

$$\left(\frac{x-\mu}{\sigma}\right)^2 = (x-\mu)(\sigma^2)^{-1}(x-\mu)$$

This is a squared statistical distance of x from ?in standard deviation units.

Generalization to $p > 1$ variables:

- We have $\mathbf{x}_{p\times 1}$ and parameters $\mu_{p\times 1}$ and $\Sigma_{p\times p}$.
- The exponent term for multivariate normal is

$$(\mathbf{x}-\mu)'\Sigma^{-1}(\mathbf{x}-\mu)$$

where $-\infty < x_i < \infty$ for $i = 1, \ldots, p$

1. This is a scalar and reduces to what's at the top for $p = 1$.
2. It is a squared statistical distance of $\mathbf{x}$ to μ (if Σ^{-1} exists). It takes into consideration both variability and covariability.
3. Integrating

$$\int_{x_1} \cdots \int_{x_p} \exp\left(-\frac{1}{2}(\mathbf{x}-\mu)'\Sigma^{-1}(\mathbf{x}-\mu)\right) = (2\pi)^{p/2}|\Sigma|^{1/2}$$

Proper Distribution

Since the sum of probabilities over all possible values must add up to 1, we need to divide by $(2\pi)^{p/2}|\Sigma|^{1/2}$ to get a "proper" density function.

Multivariate Normal density function:

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\mu)'\Sigma^{-1}(\mathbf{x}-\mu)\right)$$

where $-\infty < x_i < \infty$ for $i = 1, \ldots, p$

To denote this, we use

$$N_p(\mu, \Sigma)$$

For $p = 1$, this reduces to the univariate normal p.d.f.

Bivariate Normal Distribution

To further understand the multivariate normal distribution it is helpful to look at the bivariate normal distribution.

We have just two variables, X_1 and X_2 and that these are bivariately normally distributed with mean vector components μ_1 and μ_2 and variance-covariance matrix takes the form as shown below:

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N\left[\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix}\right].$$

Note that

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

since

$$\rho = \frac{\sigma_{12}}{\sigma_1\sigma_2}$$

Bivariate Normal

- The determinant of the variance-covariance matrix

$$|\Sigma| = \sigma_1^2\sigma_2^2(1-\rho^2)$$

- The inverse of the variance-covariance matrix

$$\Sigma^{-1} = \frac{1}{\sigma_1^2\sigma_2^2(1-\rho^2)}\begin{pmatrix} \sigma_1^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

- Therefore, the joint probability density function of (X_1, X_2) for the bivariate normal distribution as shown below:

$$f(x_1, x_2) = \frac{1}{2\pi\sqrt{\sigma_1^2\sigma_2^2(1-\rho^2)}}\exp\Big\{-\frac{1}{2(1-\rho^2)}\Big[\Big(\frac{x_1-\mu_1}{\sigma_1}\Big)^2 + \Big(\frac{x_2-\mu_2}{\sigma_2}\Big)^2 - 2\rho\Big(\frac{x_1-\mu_1}{\sigma_1}\Big)\Big(\frac{x_2-\mu_2}{\sigma_2}\Big)\Big]\Big\}$$

- If $\sigma_{12} = 0$ or equivalently $\rho = 0$, then X_1 and X_2 are uncorrelated. For bivariate normal, $\rho = 0$ implies that X_1 and X_2 are statistically independent.

Bivariate Normal& Independence

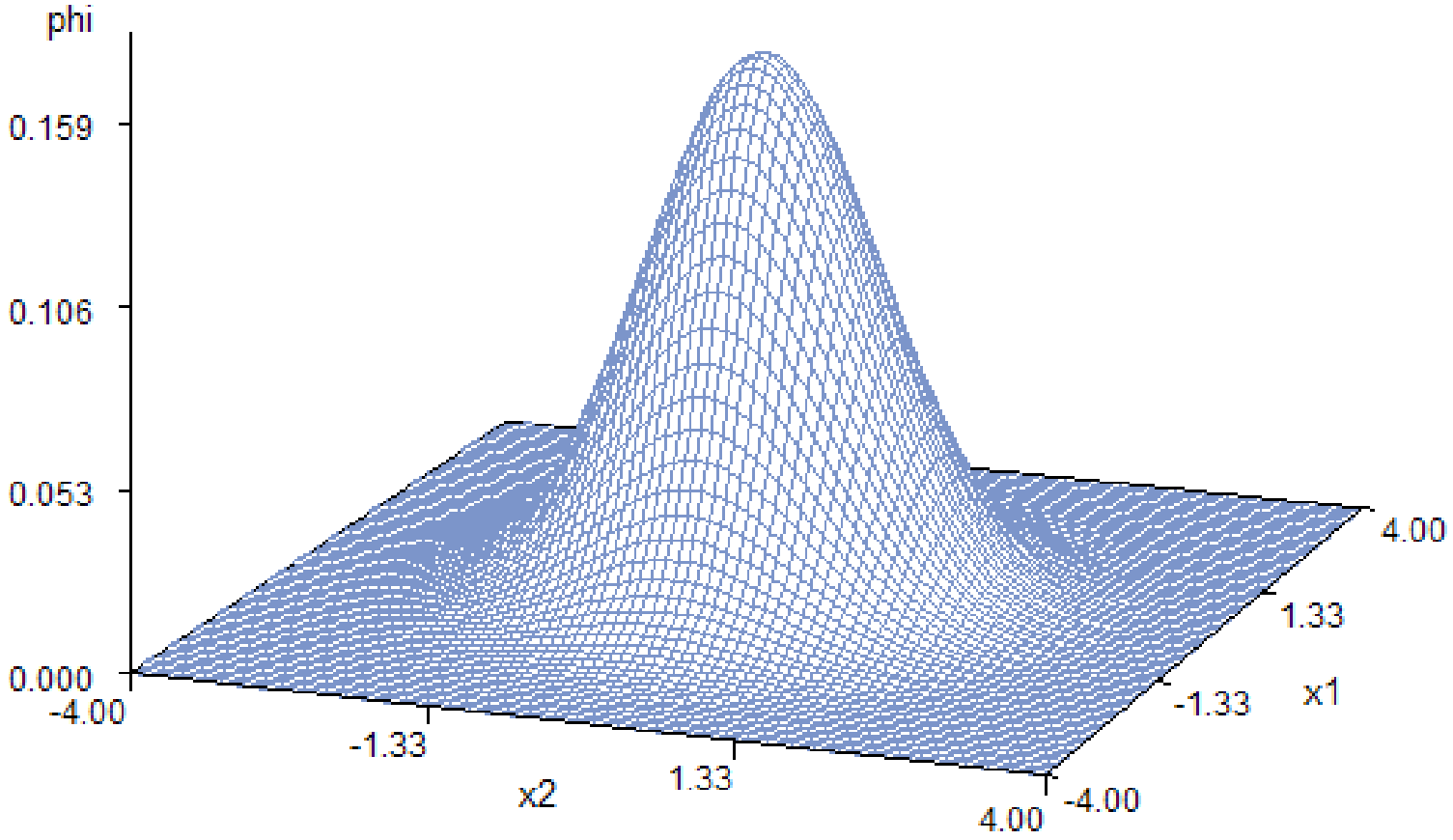
- This is because

$$
\begin{aligned}
f(x_1, x_2) &= \frac{1}{2\pi\sigma_1\sigma_2} \exp\Big\{ -\frac{1}{2}\Big[\Big(\frac{x_1-\mu_1}{\sigma_1}\Big)^2 + \Big(\frac{x_2-\mu_2}{\sigma_2}\Big)^2\Big]\Big\} \\
&= \frac{1}{\sqrt{2\pi}\sigma_1} \exp\Big\{ -\frac{1}{2}\Big(\frac{x_1-\mu_1}{\sigma_1}\Big)^2\Big\} \\
&\quad \times \frac{1}{\sqrt{2\pi}\sigma_2} \exp\Big\{ -\frac{1}{2}\Big(\frac{x_2-\mu_2}{\sigma_2}\Big)^2\Big\} \\
&= f_1(x_1) f_2(x_2)
\end{aligned}
$$

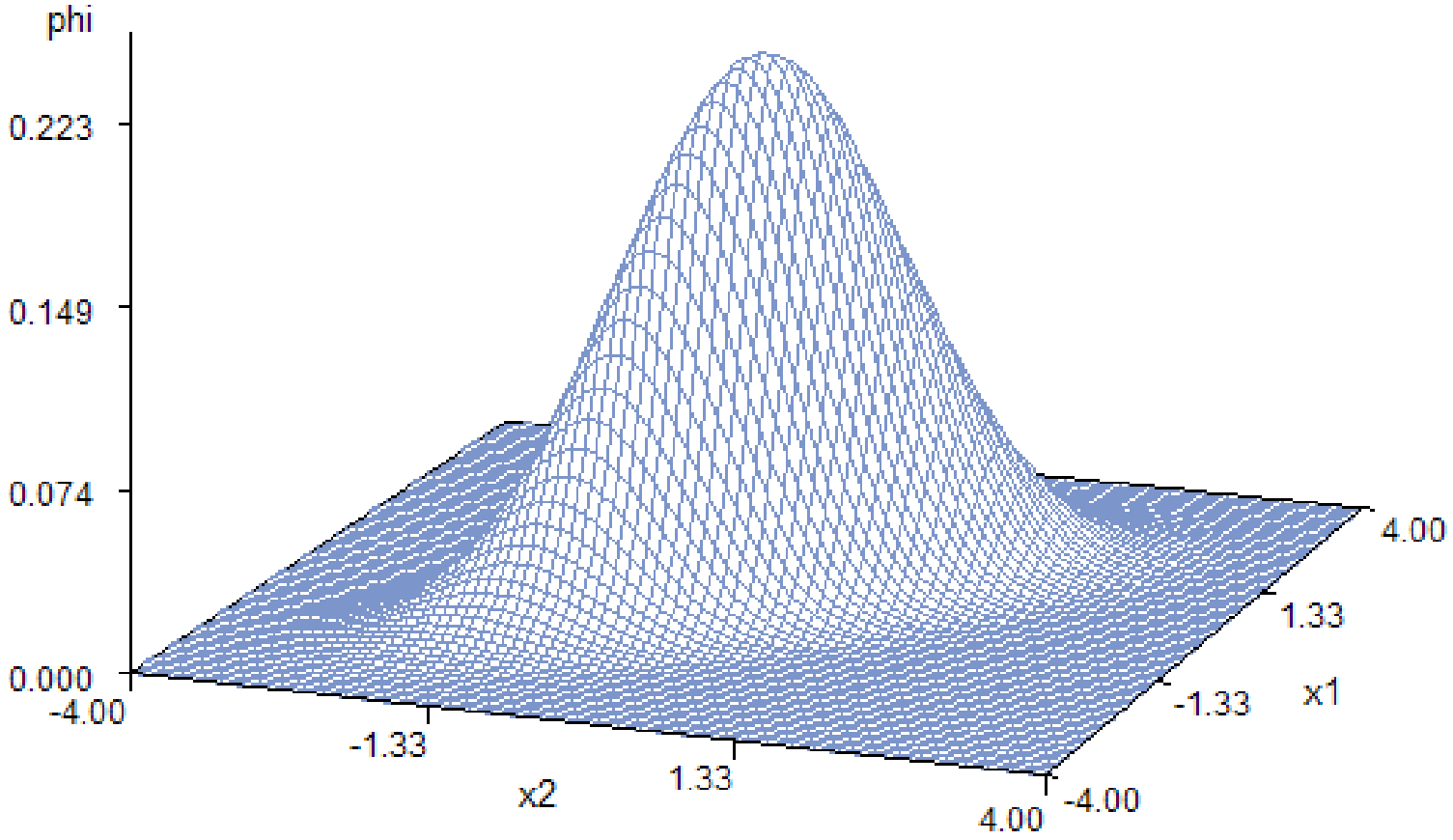
- Next we will focus on several special bivariate normal distributions with $\mu_1 = \mu_2 = 0$, $\sigma_1 = \sigma_2 = 1$ and different values of $\rho = 0, 0.7, 0.9$

$$
f(x_1, x_2) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\Big\{ -\frac{1}{2(1-\rho^2)}\Big(x_1^2 + x_2^2 - 2\rho x_1 x_2\Big)\Big\}
$$

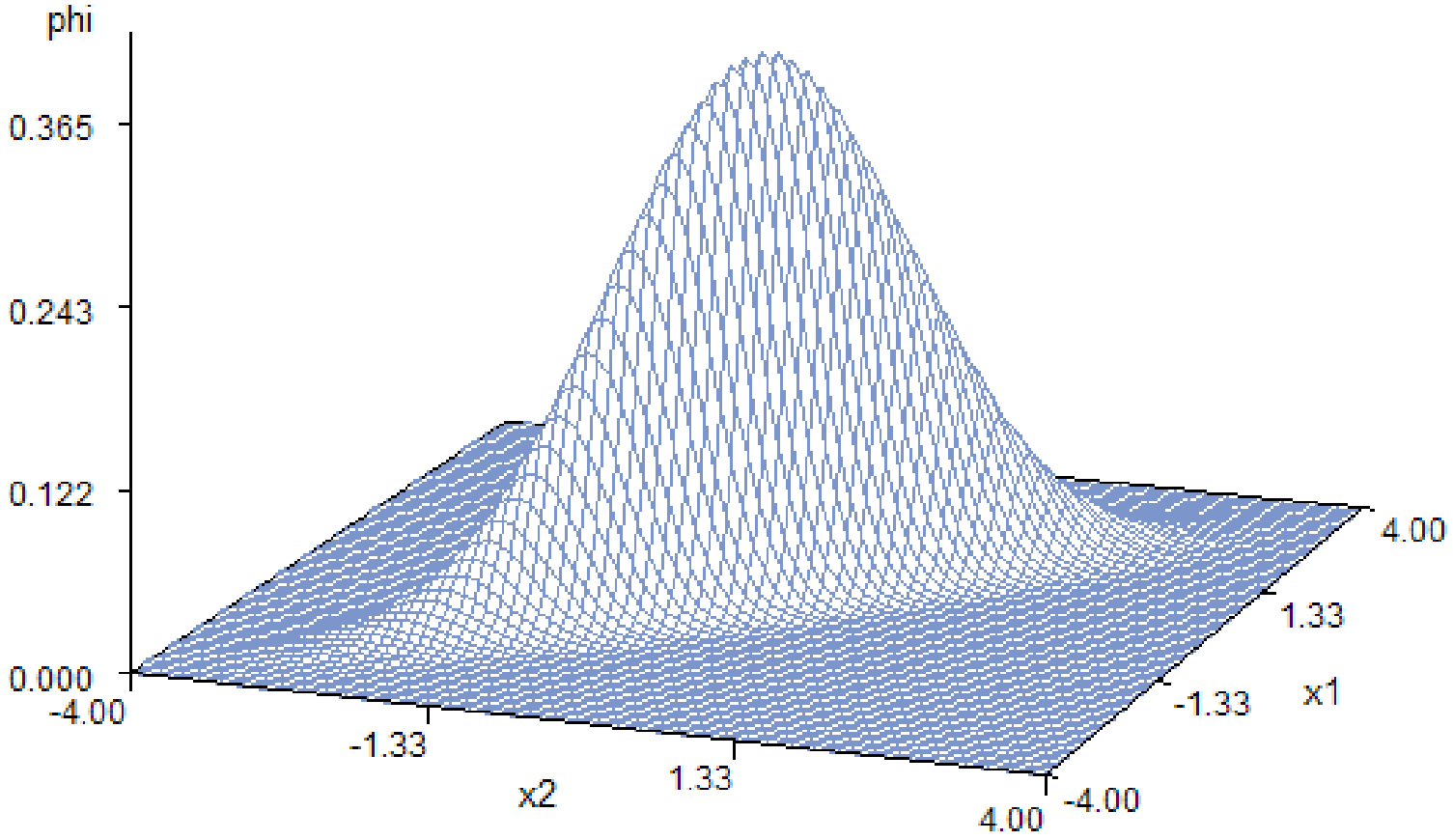
Bivariate Normal Density (r=0)



Bivariate Normal Density (r=0.7)



Bivariate Normal Density (r=0.9)



Summary of Graphs

- The first plot shows the case where the correlation ρ is equal to zero. This special case is called the circular normal distribution. Here, we have a perfectly symmetric bell-shaped curve in three dimensions.
- As ρ increases that bell-shaped curve becomes flattened on the 45 degree line. So for ρ equals 0.7 we can see that the curve extends out towards minus 4 and plus 4 and becomes flattened in the perpendicular direction.
- Increasing ρ to 0.9 the curve becomes broader and the 45 degree line and even flatter still in the perpendicular direction.

Exponent of Multivariate Normal Distribution

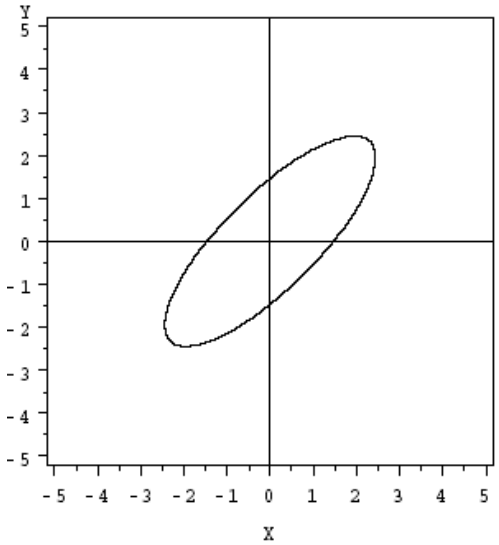
Recall the Multivariate Normal Density function below:

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\mu)'\Sigma^{-1}(\mathbf{x}-\mu)\right)$$

You will note that this density function, $f(\mathbf{x})$, only depends on $\mathbf{x}$ through the squared Mahalanobis distance:

$$(\mathbf{x}-\mu)'\Sigma^{-1}(\mathbf{x}-\mu)$$

This is the equation for a hyper-ellipse centered at μ. For a bivariate normal, where p $= 2$ variables, we have an ellipse as shown in the plot below:



Contour

A constant probability **contour** equals

$$\{\text{all } \mathbf{x} \text{ such that } (\mathbf{x}-\mu)'\mathbf{\Sigma}^{-1}(\mathbf{x}-\mu) = c^2\}$$

Useful Facts about contour:

- All values of x such that $(\mathbf{x}-\mu)'\mathbf{\Sigma}^{-1}(\mathbf{x}-\mu) = c^2$ for any specified constant value c have the same value of the density $f(\mathbf{x})$ and thus have equal likelihood.
- As the value of $(\mathbf{x}-\mu)'\mathbf{\Sigma}^{-1}(\mathbf{x}-\mu)$ increases, the value of the density function decreases. The value of $(\mathbf{x}-\mu)'\mathbf{\Sigma}^{-1}(\mathbf{x}-\mu)$ increases as the distance between $\mathbf{x}$ and μ increases.
- The variable $d^2 = (\mathbf{x}-\mu)'\mathbf{\Sigma}^{-1}(\mathbf{x}-\mu)$ has a chi-square distribution with p degrees of freedom.
- The value of d^2 for a specific observation $\mathbf{x}_j$ is called a squared Mahalanobis distance.

Probability Contours

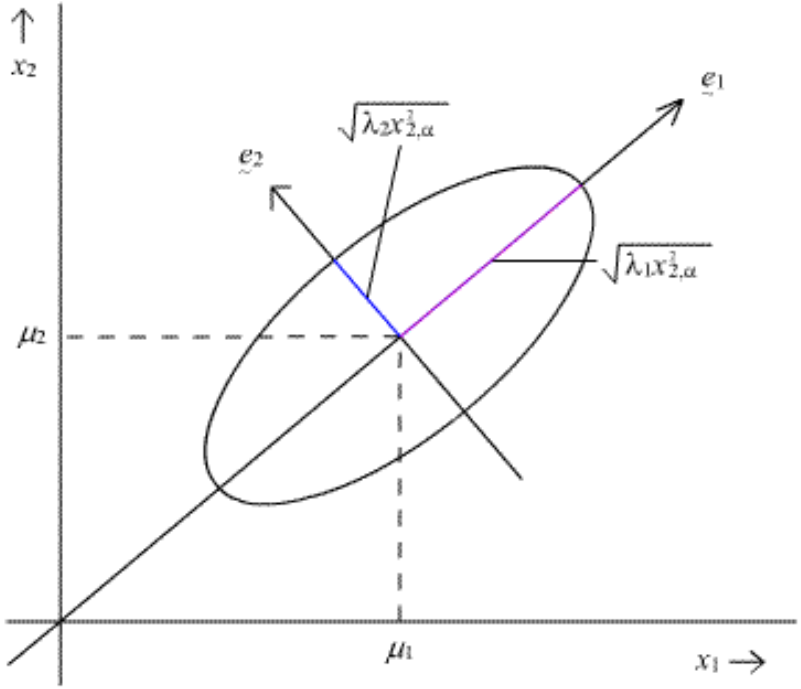
If we define a specific hyper-ellipse by taking the squared Mahalanobis distance equal to a critical value of the chi-square distribution with p degrees of freedom and evaluate this at α, then the probability that the random value X will fall inside the ellipse is going to be equal to $1 - \alpha$.

$$\Pr\{(\mathbf{x} - \mu)'\mathbf{\Sigma}^{-1}(\mathbf{x} - \mu) \leq \chi^2_{p,\alpha}\} = 1 - \alpha$$

- The axes of the ellipsoid are
  1. In the direction of the eigenvectors of Σ^{-1} (or Σ).
  2. Their lengths are proportional to the square roots of the eigenvalues of Σ.
- Specifically

$$\text{axes are} \quad \mu \pm c\sqrt{\lambda_j}\mathbf{e}_j$$

  where $\mathbf{e}_j$ and λ_j are the ith eigenvectors and eigenvalues of Σ.
- What's c?



- The $(1-\alpha)\%$ probability contour above is centered on the population means μ_1 and μ_2.
- In this diagram for the bivariate normal, the longest axis of the ellipse points in the direction of the first eigenvector $\mathbf{e}_1$ and the shorter axis is perpendicular to the first, pointing in the direction of the second eigenvector $\mathbf{e}_2$.

Example: Wechsler Adult Intelligence Scale

Here we have data on $n = 37$ subjects taking the Wechsler Adult Intelligence Test. This test is broken up into four different components:

- Information (Info)
- Similarities (Sim)
- Arithmetic (Arith)
- Picture Completion (Pic)

The data are stored in the file named wechslet.txt in five different columns. The first column is the ID number of the subjects, followed by the four component tasks in the remaining four columns.

Sample Means:

Variable	Mean
Information	12.568
Similarities	9.568
Arithmetic	11.486
Picture Completion	7.973

Variance-Covariance Matrix

$$\mathbf{S} = \begin{pmatrix} 11.474 & 9.086 & 6.383 & 2.071 \\ 9.086 & 12.086 & 5.938 & 0.544 \\ 6.383 & 5.938 & 11.090 & 1.793 \\ 2.071 & 0.544 & 1.791 & 3.694 \end{pmatrix}$$

Here, for example, the variance for Information was 11.474. For Similarities it was 12.086. The covariance between Similarities and Information is 9.086. The total variance, which is the sum of the variances comes out to be 38.344, approximately.

The eigenvalues are given below:

$$\lambda_1 = 26.245, \lambda_2 = 6.255, \lambda_3 = 3.932, \lambda_4 = 1.912$$

and finally at the bottom of the table we have the corresponding eigenvectors. They have been listed here below:

$$\mathbf{e}_1 = \begin{pmatrix} .606 \\ .605 \\ .505 \\ .110 \end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix} -.218 \\ -.496 \\ .795 \\ .274 \end{pmatrix} \mathbf{e}_3 = \begin{pmatrix} .461 \\ -.320 \\ -.335 \\ .758 \end{pmatrix}, \mathbf{e}_4 = \begin{pmatrix} -.612 \\ .535 \\ -.035 \\ .582 \end{pmatrix}$$

For example, the eigenvector corresponding the the eigenvalue 26.245, those elements are 0.606, 0.605, 0.505, and 0.110.

Now, let's consider the shape of the 95% probability contour. Note that $\chi^2_{4,.05} = 9.49$.

- First axis:

$$l_1 = \sqrt{\lambda_1 \chi^2_{4,.05}} = \sqrt{26.245 \times 9.49} = 15.782$$

  The direction of the axis is given by the first eigenvector. Looking at this first eigenvector we can see large positive elements corresponding to the first three variables. In other words, large elements for Information, Similarities, and Arithmetic. This suggests that this particular axis points in the direction specified by $\mathbf{e}_1$; that is, increasing values of Information, Similarities, and Arithmetic.

- Second axis:

$$l_2 = \sqrt{\lambda_2 \chi^2_{4,.05}} = \sqrt{6.255 \times 9.49} = 7.705$$

  Looking at the corresponding eigenvector, $\mathbf{e}_2$, we can see that this particular axis is pointed in the direction of points in the direction of increasing values for the third value, or Arithmetic and decreasing value for Similarities, the second variable.

- Third axis:

$$l_3 = \sqrt{\lambda_3 \chi^2_{4,.05}} = \sqrt{3.931 \times 9.49} = 6.108$$

  The third axis is not much shorter or smaller than the second axis. It points in the direction of $\mathbf{e}_3$; that is, increasing values of Picture Completion and Information, and decreasing values of Similarities and Arithmetic.

- Fourth axis:

$$l_4 = \sqrt{\lambda_4 \chi^2_{4,.05}} = \sqrt{1.912 \times 9.49} = 4.260$$

  It points in the direction of $\mathbf{e}_4$; that is, increasing values of Similarities and Picture Completion, and decreasing values of Information.

- The overall shape of the ellipse can be obtained by comparing the lengths of the various axis. What we have here is basically and ellipse that is the shape of a slightly squashed football.

Special Cases: $p = 2$

To further understand the shape of the multivariate normal distribution, let's return to the special case where we have $p = 2$ variables.

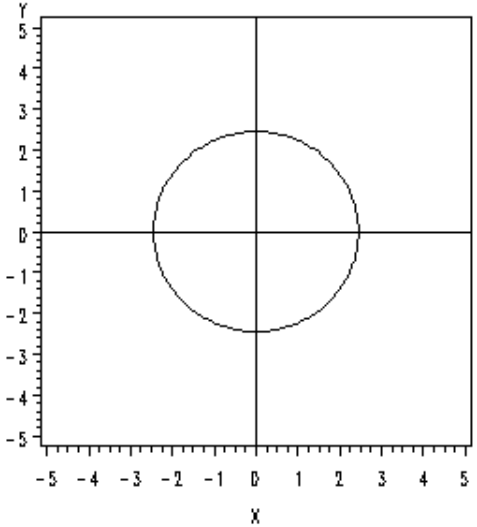
- If $\rho = 0$, there is zero correlation, then

$$\lambda_1 = \sigma_1^2, \quad \lambda_2 = \sigma_2^2$$

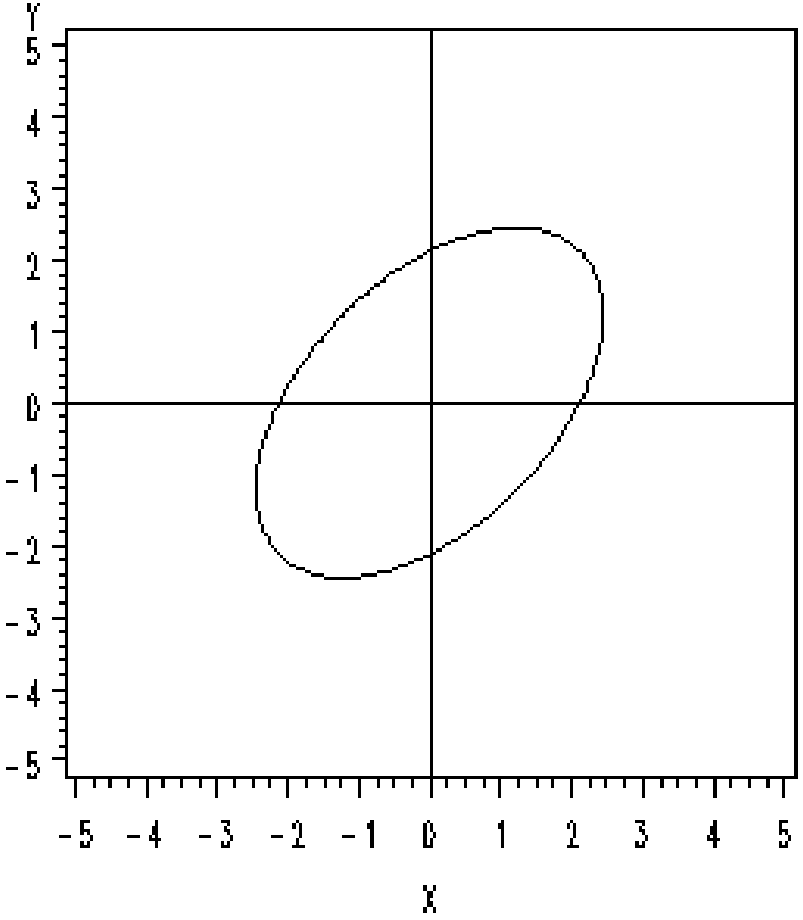
and the corresponding eigenvectors are

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

which are the same to the coordinate axis.



- If the correlation is greater than zero, then the long axis of the ellipse will have a positive slope.

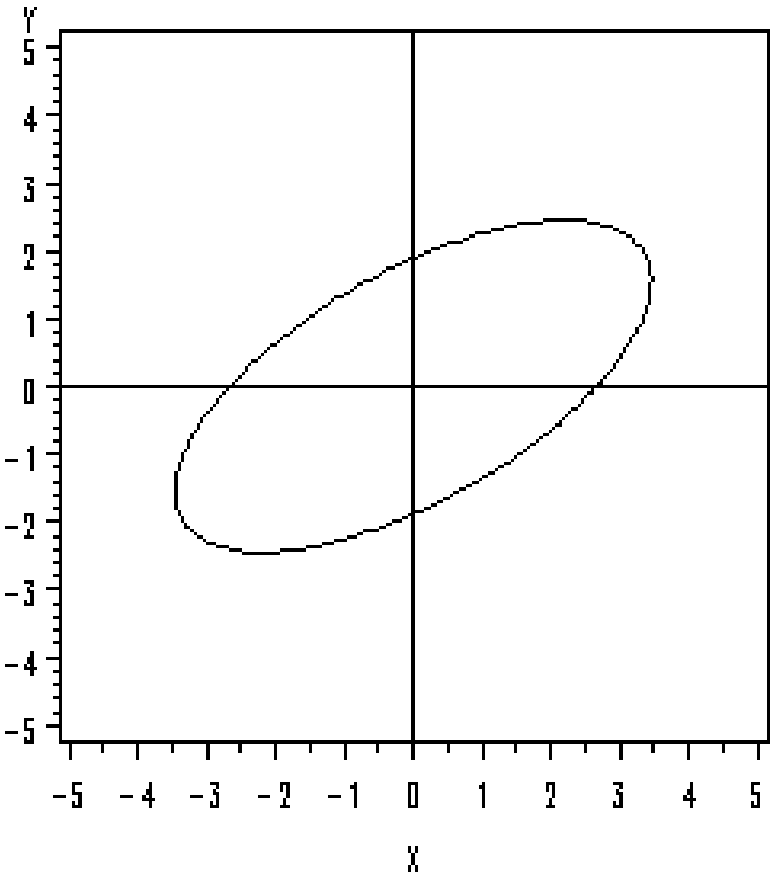


Conversely, if the correlation is less than zero, then the longest axis of the ellipse will have a negative slope.

- As the correlation approaches plus or minus 1, then largest eigenvalue will approach the sum of the two variances, and the smallest eigenvalue will approach zero:

$$\lambda_1 \to \sigma_1^2 + \sigma_2^2 \quad \text{and} \quad \lambda_2 \to 0$$

So, what is going to happen in this case is that the ellipse becomes more and more elongated as the correlation approaches one.



Plotting a 95% confidence ellipse corresponding to a specified variance-covariance matrix

More Properties that we'll Expand on

If $X \sim N_p(\mu, \Sigma)$, then

- Linear combinations of components of X are (multivariate) normal.
- All sub-sets of the components of X are (multivariate) normal.
- Zero covariance implies that the corresponding components of X are statistical independent.
- The conditional distributions of the components of X are (multivariate) normal.

1: Linear Combinations

If $X \sim N_p(\mu, \Sigma)$, then any linear combination

$$\mathbf{a}'X = a_1X_1 + a_2X_2 + \ldots + a_pX_p$$

is distributed as

$$\mathbf{a}'X \sim N(\mathbf{a}'\mu, \mathbf{a}'\Sigma\mathbf{a})$$

Also, If $\mathbf{a}'X$ is normal $N(\mathbf{a}'\mu, \mathbf{a}'\Sigma\mathbf{a})$ for all possible $\mathbf{a}$, then X must be $N_p(\mu, \Sigma)$.

Example:

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N_2\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix}\right)$$

Let $Y = X_1 + 2X_2$, find the distribution of Y.

More Linear Combinations

If $X \sim N_p(\mu, \Sigma)$, then the q linear combination

$$Y_{q\times 1} = \mathbf{A}_{q\times p} X_{p\times 1} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1p} \\ a_{21} & a_{22} & \dots & a_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ a_{q1} & a_{q2} & \dots & a_{qp} \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{pmatrix}$$

is distributed as $N_q(\mathbf{A}\mu, \mathbf{A}\Sigma\mathbf{A}')$.

Example:

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N_2\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix}\right)$$

Let

$$Y = \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = \begin{pmatrix} X_1 + 2X_2 \\ 2X_1 - X_2 \end{pmatrix},$$

find the distribution of Y.

2: Sub-sets of Variables

If $X \sim N_p(\mu, \Sigma)$, then all sub-sets of X are (multivariate) normally distributed.

For example, let's partition X into two sub-sets

$$X_{p\times 1} = \begin{pmatrix} X_1 \\ \vdots \\ X_q \\ \hline X_{q+1} \\ \vdots \\ X_p \end{pmatrix} = \begin{pmatrix} X^{(1)}_{q\times 1} \\ \hline X^{(2)}_{(p-q)\times 1} \end{pmatrix}, \mu = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_q \\ \hline \mu_{q+1} \\ \vdots \\ \mu_p \end{pmatrix} = \begin{pmatrix} \mu^{(1)}_{q\times 1} \\ \hline \mu^{(2)}_{(p-q)\times 1} \end{pmatrix}$$

$$\Sigma = \left(\begin{array}{c|c} \Sigma^{11}_{q\times q} & \Sigma^{12}_{q\times(p-q)} \\ \hline \Sigma^{21}_{(p-q)\times p} & \Sigma^{22}_{(p-q)\times(p-q)} \end{array}\right) = \left(\begin{array}{c|c} \Sigma_{11} & \Sigma_{12} \\ \hline \Sigma_{21} & \Sigma_{22} \end{array}\right)$$

Then

$$X^{(1)} \sim N_q(\mu^{(1)}, \Sigma_{11}) \quad \text{and} \quad X^{(2)} \sim N_{(p-q)}(\mu^{(2)}, \Sigma_{22})$$

3. Zero Covariance & Statistical Independence

- If X_1 is $(q_1 \times 1)$ and X_2 is $(q_2 \times 1)$ are statistically independent, then $\text{cov}(X_1, X_2) = \Sigma_{12} = 0$.
- If
$$\left(\frac{X_1}{X_2}\right) \sim N_{q_1+q_2}\left(\left(\frac{\mu_1}{\mu_2}\right), \left(\begin{array}{c|c} \Sigma_{11} & \Sigma_{12} \\ \hline \Sigma_{21} & \Sigma_{22} \end{array}\right)\right),$$
Then X_1 and X_2 are statistically independent if and only if $\Sigma_{12} = \Sigma_{21} = 0$.
- If X_1 and X_2 are statistically independent and distributed as $N_{q_1}(\mu_1, \Sigma_{11})$ and $N_{q_2}(\mu_2, \Sigma_{22})$, respectively, then
$$\left(\frac{X_1}{X_2}\right) \sim N_{q_1+q_2}\left(\left(\frac{\mu_1}{\mu_2}\right), \left(\begin{array}{c|c} \Sigma_{11} & \mathbf{0} \\ \hline \mathbf{0} & \Sigma_{22} \end{array}\right)\right),$$

4: Conditional Distributions

Let

$$\left(\frac{X_1}{X_2}\right) \sim N_{q_1+q_2}\left(\left(\frac{\mu_1}{\mu_2}\right), \left(\begin{array}{c|c} \Sigma_{11} & \Sigma_{12} \\ \hline \Sigma_{21} & \Sigma_{22} \end{array}\right)\right),$$

and $|\Sigma| > 0$, (i.e. positive definite), Then the conditional distribution of X_1 given $X_2 = \mathbf{x}_2$ is (multivariate) normal with mean and covariance matrix

$$\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \mu_2) \quad \text{and} \quad \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

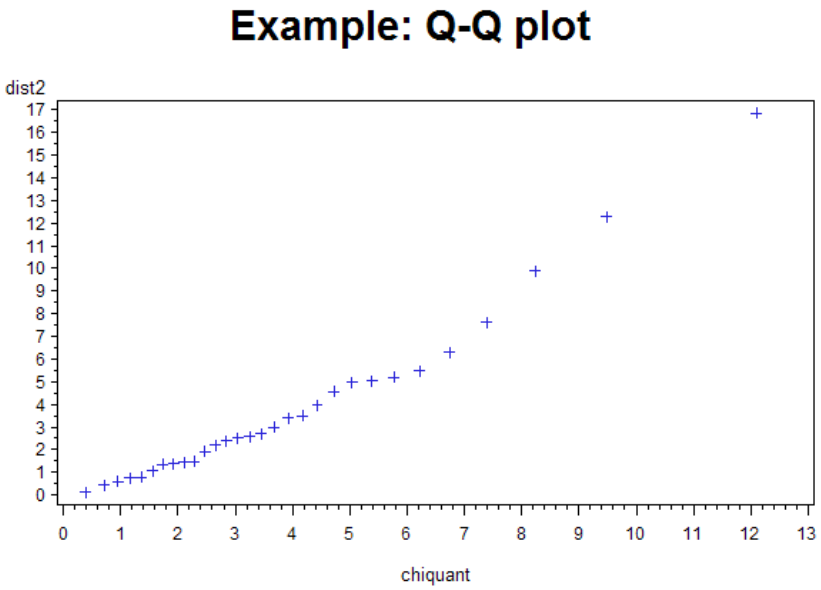
Q-Q Plot for Multivariate Normality and Outliers

- The variable $d^2 = (\mathbf{x} - \mu)'\mathbf{\Sigma}^{-1}(\mathbf{x} - \mu)$ has a chi-square distribution with p degrees of freedom, and for "large" samples the observed Mahalanobis distances have an approximate chi-square distribution. This result can be used to evaluate (subjectively) whether a data point may be an outlier and whether observed data may have a multivariate normal distribution.
- A Q-Q plot can be used to picture the Mahalanobis distances for the sample. The basic idea is the same as for a normal probability plot. For multivariate data, we plot the ordered Mahalanobis distances versus estimated quantiles (percentiles) for a sample of size n from a chi-squared distribution with p degrees of freedom. This should resemble a straight-line for data from a multivariate normal distribution. Outliers will show up as points on the upper right side of the plot for which the Mahalanobis distance is notably greater than the chi-square quantile value.

Q-Q plot

- Determining the Quantiles: The ith estimated quantile is determined as the chi-square value (with df $= p$) for which the cumulative probability is $(i - 0.5)/n$. To determine the full set of estimated chi-square quantiles, this is done for value of i from 1 to n.
- Example: Q-Q Plot for Board Stiffness Data: This example reproduces Example 4.14 in the text (page 187) . For each of n = 30 boards, there are p = 4 measurements of board stiffness. Each measurement was done using a different method.

Q-Q plot



The distances are on the vertical and the chi-square quantiles are on the horizontal. At the right side of the plot we see an upward bending. This indicates possible outliers (and a possible violation of multivariate normality). In particular, the final point has $d^2 \approx 16$ whereas the quantile value on the horizontal is about 12.5. The next-to-last point on the plot might also be an outlier. A printout of the distances, before they were ordered for the plot, shows that the two possible outliers are boards 16 and 9, respectively.

Calculating Mahalanobis Distance With SAS

SAS does not provide Mahalanobis distance directly, but we can compute them using principal components. The steps are:

- Determine principal components for the correlation matrix of the x-variables.
- Standardize the principal component scores so that each principal component has standard deviation = 1. For each component, this is done by dividing the scores by the square root of the eigenvalue. In SAS, use the STD option as part of the PROC PRINCOMP command to automate this standard deviation. For each observation, calculate d2 = sum of squared standardized principal components scores. This will equal the squared Mahalanobis distance.
- For each observation, calculate d^2 = sum of squared standardized principal components scores. This will equal the squared Mahalanobis distance.

Example: Mahalanobis Distance

```
PROC PRINCOMP std out=pcout;
var x1 x2 x3 x4;
run;
DATA out=mahal;
set pcout;
dist2 =uss(of prin1-prin4);
run;
```

Suppose we have four x-variables, called $x1, x2, x3, x4$, and they have already been read into SAS. The above SAS code (mahalonobis.sas) will determine standardized principal components and calculate Mahalanobis distances (the printout will include observation numbers). Within the DATA step, the "uss(of prin1-prin4)" function calculates the uncorrected sum of squares for the variables prin1-prin4. This value will be computed for each observation in the "pcout" data set. The result of the DATA step will be a SAS data set named "mahal" that will include the original variables, the standardized principal component scores (named prin1-prin4) and the Mahalanobis distance (named dist2).

Example: code for Q-Q plot

- The data step reads the dataset.
- The proc princomp calculates the principal components and stores the standardized principal components in a dataset named pcresult.
- The next data step calculates the Mahalanobis distances and keeps them in a dataset named mahal.
- The proc print will print the distances (with observation numbers).
- Next we sort the mahal dataset in order of the distances. This is to prepare for the Q-Q plot.
- In the next data step we compute estimated quantiles of a chi-square distribution with df $= 4$. In the prb $=$ line, the value 30 is the sample size and in the cinv function the value 4 is the df (because we have 4 variables).